Layer 15: Calibration and Clinical Interpretation

Single-Channel MMP-9 Calibration, Detection Limits, Interference, Matrix Correction, and Bayesian Wound-State Classification

Eton iGEM 2026, Dry Lab | Layer 15 of 17-Layer Biosensor Pipeline | `calibration_curve.py` · `deconvolution.py` · `matrix_correction.py` · `wound_classifier.py`

Layer 15 is the inverse-problem layer of the device model chain: given a measured fractional cleavage F_k on the fabricated electrode, what is the wound-fluid MMP-9 concentration, and what does it imply clinically? It takes the surface kinetics of Layer 9 and the ratiometric signal chain of Layer 13 as fixed inputs and builds four downstream tools: a calibration-curve fit with detection limits, an interference-bias check for the one significant off-target protease (HNE), a matrix-effect / spike-and-recovery correction, and a Bayesian wound-state classifier.

Architecture note. Earlier planning material (`contents.md`, Section 15.5) specified a two-probe device (separate MMP-9 and HNE channels) with a 2×2 cross-sensitivity matrix S and a condition-number design criterion $\kappa(S) < 30$. Layers 9 and 13 confirm the fabricated design is not this: it is a *single* MMP-9 channel (active electrode: SGKIPRTLTA; negative control: the D-enantiomer of the same peptide, a stereospecific control – Layer 13, Sec. III). There is no HNE-specific probe. This report therefore supersedes the two-channel framing: HNE is treated throughout as a possible *interferent* on the single MMP-9 channel (Layer 9, Sec. X), not as a second measurable channel.

I. Calibration Model

The progress-curve equation, and a units correction

Layer 9 (Sec. IV) derives the surface Michaelis–Menten progress curve for fractional probe cleavage F_k as an implicit function of the free enzyme concentration $[E]$, the incubation time T, the surface rate constant $k_{\text{surf}} \equiv k_{\text{cat}}/K_m^{\text{surf}}$ ($\text{M}^{-1}\text{s}^{-1}$), and the probe-saturation ratio $\rho = \Gamma_0/K^{\Gamma}_{m,\text{surf}}$ (Layer 9, Eq. IV.4). As stated in Layer 9 (Eq. IV.5), the equation reads $k_{\text{cat}}[E]t = K^{\Gamma}_{m,\text{surf}} \ln\frac{1}{1-F_k} + \Gamma_0 F_k$. Implementing this directly exposed a dimensional inconsistency: the left side is dimensionless (k_{surf} in $\text{M}^{-1}\text{s}^{-1} \times [E]$ in M $\times\, t$ in s), while the right side carries units of mol cm^{-2} ($K^{\Gamma}_{m,\text{surf}}, \Gamma_0 \sim 10^{-13}$–$10^{-12}$) – a mismatch of roughly twelve orders of magnitude. Layer 9's own Eq. IV.2 converts the solution-phase K_m into areal units via a probe-layer thickness δ_{eff}, but that conversion is not carried through consistently into Eq. IV.5.

Dividing Layer 9's Eq. IV.5 through by $K^{\Gamma}_{m,\text{surf}}$ gives a dimensionally consistent equation that (a) uses Layer 9's own dimensionless ratio ρ, and (b) exactly reproduces Layer 9's stated probe-dilute limit ($\rho \to 0$: $F_k = 1 - \exp(-k_{\text{surf}}[E]T)$, their Eq. IV.7). This corrected form is used throughout Layer 15:

$$k_{\text{surf}}\,[E]\,T \;=\; \ln\frac{1}{1-F_k} \;+\; \rho\,F_k \tag{I.1}$$

where $\rho = \Gamma_0/K^{\Gamma}_{m,\text{surf}}$. Using Layer 9's own reference values ($K^{\Gamma}_{m,\text{surf}} = 6.4 \times 10^{-13}$ mol cm^{-2}, from an unconfirmed $\delta_{\text{eff}} = 5$ nm; $\Gamma_0 = 2.0 \times 10^{-12}$ mol cm^{-2}, the Pheeney 2012 midpoint), $\rho \approx 3.1$, matching Layer 9's own worked example (Table 1, "no correction" row) – well outside the probe-dilute limit, so the implicit form is required rather than a simple exponential.

Forward and inverse solutions

Equation I.1 is monotonic in $F_k \in [0, 1)$ for any fixed $[E]$, so the forward direction ($[E] \to F_k$) is solved by bisection (`scipy.optimize.brentq`). The inverse direction ($F_k \to [E]$, i.e. the actual quantification

step) is closed-form and needs no solver:

$$[E] \;=\; \frac{\ln\frac{1}{1-F_k} + \rho\, F_k}{k_{\text{surf}}\, T} \tag{I.2}$$

Fitting k_{surf} from calibration data

Only one parameter is fit from a calibration curve: k_{surf}. ρ is held fixed at the Layer 9 estimate (itself unconfirmed pending a measured δ_{eff} from Layer 8), and $T = 1800$ s by protocol. The solution-phase value $k_{\text{cat}}/K_m = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ is **confirmed** for SGKIPRTLTA (Kridel 2001, Table II) and is an *upper bound* on the surface value; Layer 9 (Sec. XIII) states surface immobilisation is expected to reduce this 2–100×, giving a plausible surface range of 2,000–67,500 $\text{M}^{-1}\text{s}^{-1}$, with 13,000–27,000 $\text{M}^{-1}\text{s}^{-1}$ consistent with a 30–60 min read window (Layer 9, Table 2). Pending fabrication, calibration curves in this report are simulated with a synthetic placeholder $k_{\text{surf}} = 2.0 \times 10^4\ \text{M}^{-1}\text{s}^{-1}$, chosen within that target range, and refit from the simulated data as a self-consistency check (Fig. 1).

II. Uncertainty Model

Layer 13 (Sec. III, Eq. III.6) derives the propagated uncertainty on F_k from the full ratiometric error chain: peak-area fit noise $\rightarrow \sigma_R/R \rightarrow \sigma_\rho/\rho \rightarrow \sigma_{F_k}$. The result is *heteroscedastic* – it depends on F_k itself, not a flat blank-noise floor:

$$\sigma_{F_k}(F_k) \;=\; (1 - F_k)\,\sqrt{2}\,\frac{\sigma_\rho/\rho}{\sqrt{N_{\text{active}}}} \tag{II.1}$$

with $\sigma_\rho/\rho \approx 4.8\%$ per active/control electrode pair on a single scan (Layer 13, Eq. II.6), reducible by $1/\sqrt{N_{\text{active}}}$ when averaging over N_{active} paired electrodes. At $F_k = 0$, single electrode: $\sigma_{F_k}(0) \approx 6.8\%$ – roughly 30× larger than an earlier flat 0.2% placeholder used before Layers 9/13 were available, and the reason Layer 13 itself sets its detection floor at $F_k = 20\%$ rather than a few percent (Sec. IV, below).

III. Detection Limits and Dynamic Range

Layer 13 (Sec. IV) defines the usable window directly on F_k: $F_{\min} = 0.20$ ($\approx 3\sigma_{F_k}$ at $F_k = 0$, single electrode) and $F_{\max} = 0.95$ (beyond this the MB signal is indistinguishable from noise and Eq. I.2's prefactor diverges). This differs from a generic IUPAC $3\sigma/10\sigma$ blank-slope LOD/LLOQ definition (ICH Q2(R1), 2005 – see References) – Layer 13's own thresholds are used here instead, since they are already derived from the device's real noise model rather than a linear-regime approximation.

$[E]$-space limits follow directly from Eq. I.2 at $F_{\min}$ and $F_{\max}$: $\text{LOD} = [E](F_{\min})$, $\text{ULOQ} = [E](F_{\max})$.

Result, at the synthetic placeholder k_{surf}

Fitting the simulated calibration curve (Fig. 1) recovers $k_{\text{surf}} = 1.97 \times 10^4 \pm 6.5 \times 10^2\ \text{M}^{-1}\text{s}^{-1}$ (input: 2.0×10^4), giving:

Quantity	Value
LOD ($F_k = 20\%$)	23.9 nM
ULOQ ($F_k = 95\%$)	168.2 nM
Dynamic range	0.85 decades
Covers clinical range (6–30 nM)?	**No**

This is a genuine finding, not a bug: at this placeholder k_{surf}, the device's LOD (23.9 nM) sits *above* the confirmed healing-wound concentration (6 nM, Layer 0/9), so a healing-range signal would currently fall below the detection floor. Section VIII revisits this against the wound classifier's decision boundary. As a cross-check, Layer 13's own closed-form window on the apparent rate $\kappa_k = -\ln(1 - F_k)/T$ (Eq. IV.5) gives $[\kappa_{\min}, \kappa_{\max}] = [1.2 \times 10^{-4},\ 2.0 \times 10^{-3}]\ \text{s}^{-1}$, a $13.4\times$ (1.13-decade) window – consistent in scale with the $[E]$-space result above, though not identical, since the two use different (rate- vs. concentration-based) definitions.

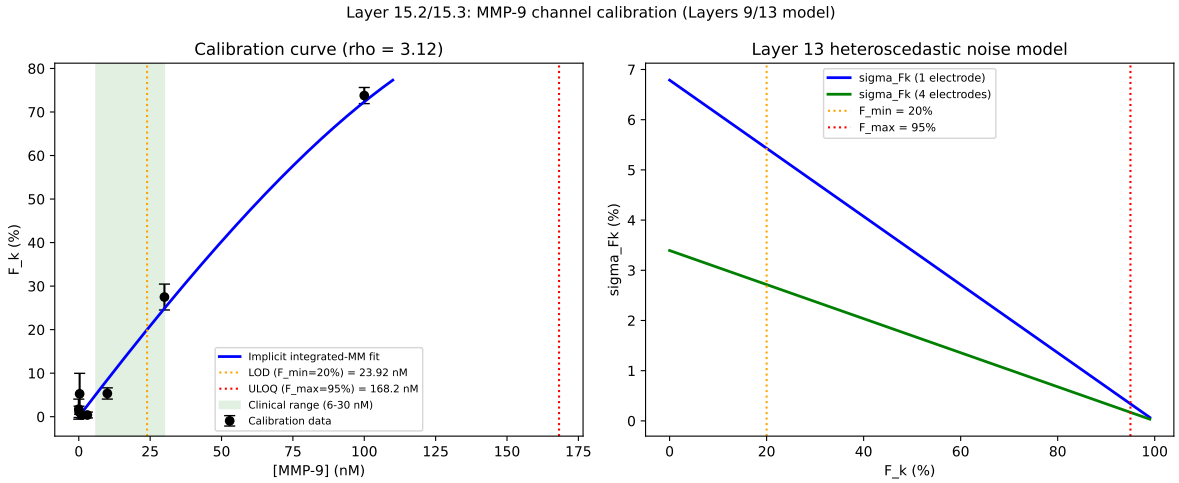


Figure 1: Left: simulated calibration data and the fitted implicit-MM curve (Eq. I.1), with LOD, ULOQ, and the confirmed clinical range (6–30 nM, Layer 0/9) marked. Right: the heteroscedastic noise model (Eq. II.1) at 1 and 4 averaged electrodes.

IV. HNE Interference on the Single Channel

With no second probe, HNE cannot be deconvolved from MMP-9 – it can only bias the single MMP-9 channel if it cross-cleaves SGKIPRTLTA at all. Layer 9 (Sec. X, Table 3) flags this as unconfirmed with "moderate risk" and gives no measured cross-reactivity constant. In the first-order regime, rate contributions from different proteases on the same probe add (Layer 9, Eq. X.1):

$$k_{\text{obs,total}} = k_{\text{surf}}^{\text{MMP9}}[\text{MMP9}] + k_{\text{surf}}^{\text{HNE}}[\text{HNE}], \qquad \xi_{\text{HNE}} = \frac{k_{\text{surf}}^{\text{HNE}}[\text{HNE}]}{k_{\text{obs,total}}} \tag{IV.1}$$

Because the device inverts F_k assuming pure MMP-9 signal (Eq. I.2), any nonzero ξ_{HNE} biases the inferred $[\text{MMP9}]$ upward. Using a 3% cross-reactivity placeholder ($k_{\text{surf}}^{\text{HNE}} = 0.03\, k_{\text{surf}}^{\text{MMP9}}$, pending measurement) at the confirmed clinical concentrations (chronic: 30 nM MMP-9 + 30 nM HNE; healing: 6 nM MMP-9 + 0 nM HNE, Layer 0):

State	F_k observed	ξ_{HNE}	Apparent [MMP-9] (bias)
Chronic	26.0%	2.9%	30.9 nM (+3.0%)
Healing	5.2%	0.0%	6.00 nM (−0.0%)

giving an apparent chronic/healing ratio of $5.15\times$ against a clean (no-interference) value of $5.0\times$ (Layer 9, Eq. VIII.3) – a small perturbation at the 3% placeholder. Sweeping the cross-reactivity fraction (Fig. 2)

shows the bias grows steeply: it crosses 20% (comparable to the LOD-scale detection floor) at roughly 15–20% cross-reactivity, well within the range Layer 9 itself leaves open as possible. Measuring the true HNE-on-SGKIPRTLTA rate constant is therefore a priority alongside $k_{\text{surf}}^{\text{MMP9}}$ itself.

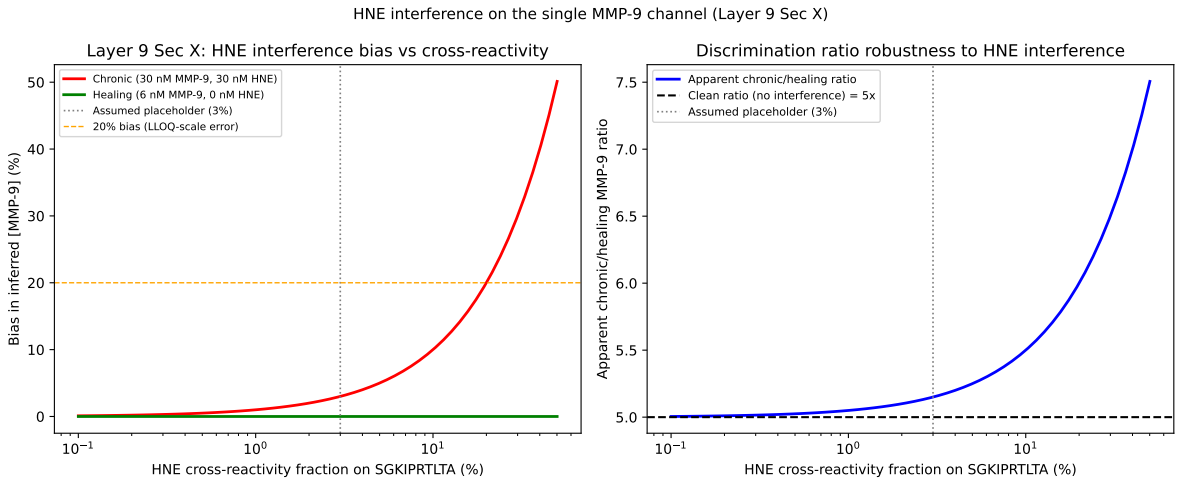


Figure 2: Left: bias in inferred [MMP-9] vs. assumed HNE cross-reactivity fraction, at chronic and healing clinical concentrations. Right: the resulting apparent chronic/healing discrimination ratio vs. the clean 5× value.

V. Matrix Effects and Spike-and-Recovery

What the double ratio already cancels

Layer 13's D-enantiomer double ratio, $F_k = 1 - \rho_{\text{active}}/\rho_{\text{control}}$, already cancels *exactly* any non-stereospecific background cleavage and *uniform* electrode fouling (Layer 13, Sec. II–III). A naive matrix-correction model that re-litigates those two effects would be redundant. What Layer 13 itself flags as *not* cancelled (Sec. VII, Limitation 2) is *differential* fouling between the active and control electrodes – so this is what a matrix-correction model should actually target, together with a genuine shift in enzyme catalytic efficiency (k_{surf}) from ionic strength, Ca^{2+}, and protein crowding in real wound-fluid-matched matrix, and pH (Sec. 15.6, `contents.md`).

Recovery model

Spiked matrix cartridges are simulated with $k_{\text{surf}}^{\text{matrix}} = k_{\text{surf}}^{\text{nominal}} \times a_{\text{matrix}} \times \text{pH_correction}(\text{pH})$, plus a small residual differential-fouling bias on F_k (placeholder 1.5%, unconfirmed). Recovery is defined against the buffer-only nominal calibration curve:

$$\text{Recovery}(\%) = \frac{F_{k,\text{spiked}} - F_{k,\text{blank}}}{F_{k,\text{calibration}}} \times 100 \tag{V.1}$$

At a 15% matrix attenuation placeholder ($a_{\text{matrix}} = 0.85$) and pH 7.4, mean recovery at mid-to-high spiked concentrations is 83.8% ($m = 1.19\times$ correction factor) – within the 80–120% acceptance band (`contents.md`, Sec. 15.4) and consistent with the input attenuation. Near-LOD spike points (1–3 nM) are excluded from the mean: noise in F_k (Eq. II.1) is amplified by dividing by a small calibration F_k, producing recovery estimates that swing from -200% to $+300\%$ at trace levels (Fig. 3) – an expected artefact of the ratio, not a real matrix effect, and the reason real spike-and-recovery protocols avoid trace-level spikes.

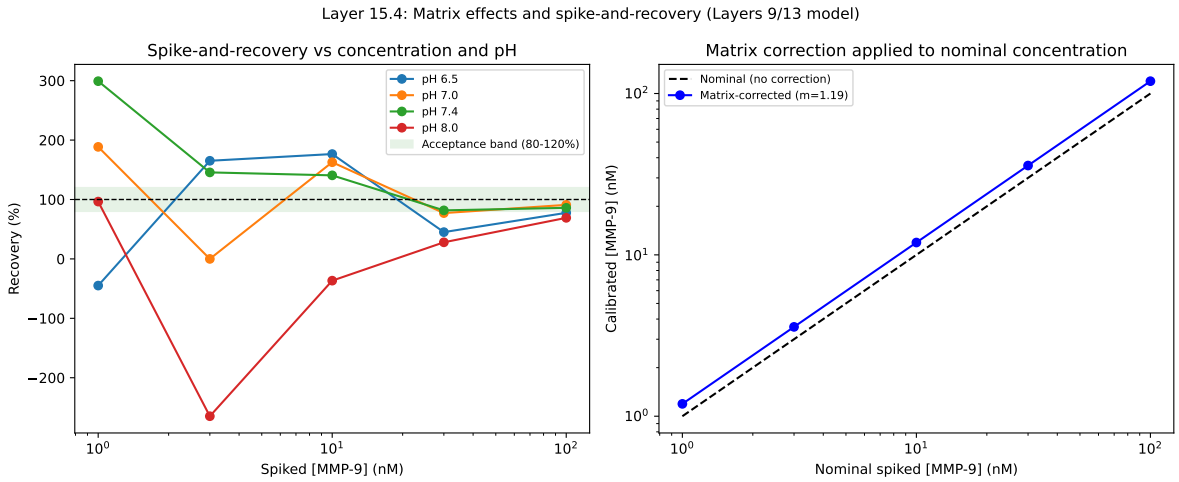


Figure 3: Left: simulated spike-and-recovery vs. concentration and pH; low-concentration points are noise-dominated. Right: matrix-corrected vs. nominal concentration axis using the fitted correction factor m.

VI. Bayesian Wound-State Classification

Two states, not three

Layer 0 analyses only Healing and Chronic wounds; no literature cited anywhere in Layers 0/9/13 defines or supports a third "Inflammatory" intermediate category, so the classifier here uses two states.

Primary classifier: MMP-9 alone

Because the fabricated device has a single MMP-9 channel, the clinically realistic classifier conditions on $[\mathrm{MMP9}]$ alone. Confirmed values from Layer 0's full ODE solution at $t = 30$ min are $[\mathrm{MMP9}]_{\mathrm{chronic}} = 31.9$ nM and $[\mathrm{MMP9}]_{\mathrm{healing}} = 6.2$ nM (rounded to 30 and 6 nM respectively in Layer 9 and used identically here for consistency with the calibration numbers above). Likelihoods are log-normal, with a common $\sigma_{\log_{10}} = 0.15$ ($\approx 40\%$ relative) for both states: Layer 0 characterises only the healing estimate as "order-of-magnitude" (Sec. II) and gives no formally propagated uncertainty for either state, so a single conservative symmetric value is used rather than an invented asymmetric split.

$$P(\text{state} \mid [\mathrm{MMP9}]) \;\propto\; \mathcal{N}\big(\log_{10}[\mathrm{MMP9}];\; \mu_{\text{state}},\; \sigma^2_{\log_{10}}\big)\, P(\text{state}) \tag{VI.1}$$

with equal priors $P(\text{state}) = 0.5$. The maximum-a-posteriori decision boundary ($P(\text{Healing}) = P(\text{Chronic})$) is at $\mathbf{13.4}$ nM (Fig. 4).

Illustrative secondary model: MMP-9 + HNE

A bivariate extension is retained only to show what a hypothetical second channel would add. Layer 0's own conclusion undercuts the case for building one: HNE is elevated in only 6 of 13 chronic samples (Trengove 1999) – roughly half of chronic wounds are HNE-low regardless of state – so Layer 0 explicitly frames HNE as "not a reliable standalone discriminator" and "a secondary reference channel, not a primary diagnostic output." Consistent with this, adding HNE moves the chronic-state posterior negligibly (from 100.0% to 100.0% across HNE = 1–30 nM at fixed MMP-9 = 31.9 nM in the simulated check): the MMP-9 axis alone is already doing essentially all of the discriminating work.

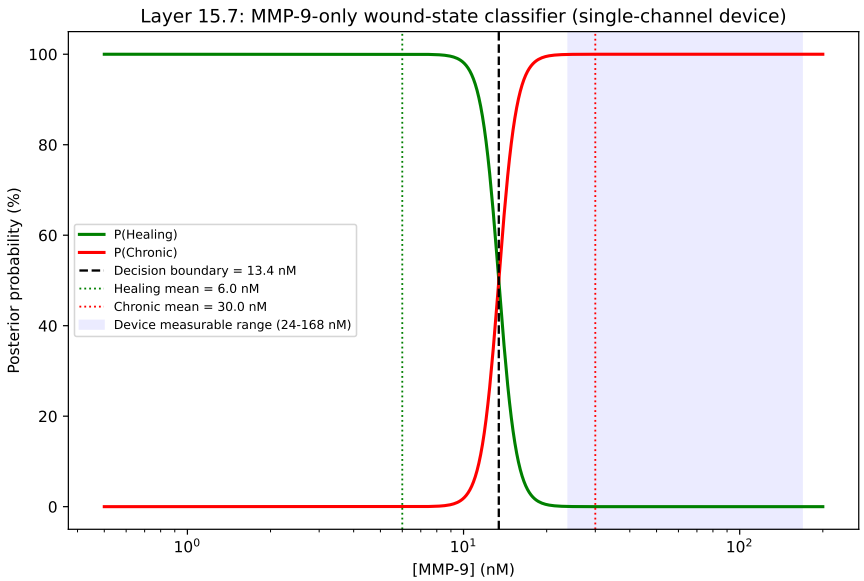


Figure 4: Posterior probability vs. [MMP-9] for the primary (MMP-9-only) classifier, with the decision boundary, confirmed state means, and the device's own measurable range (Section III) overlaid.

VII. Cross-Layer Integration Check

Overlaying the classifier's decision boundary on the calibration curve's own measurable range (Fig. 4) surfaces the report's central finding: the boundary (13.4 nM) falls *below* the LOD (23.9 nM) at the current placeholder k_{surf}. In other words, at this (unmeasured, synthetic) surface rate constant, the device cannot yet reliably resolve concentrations near the clinically decisive threshold – it is blind exactly where the classification depends most sensitively on signal. This is not a flaw in the classifier or the calibration model individually; it only appears by running both together, and it is the clearest concrete argument in this report for prioritising a real k_{surf} measurement on the fabricated electrode: a higher true k_{surf} (within Layer 9's own 13,000–27,000 $\text{M}^{-1}\text{s}^{-1}$ target band) would push the LOD down and could resolve this gap without any change to the classifier itself.

VIII. Key Outputs

Output	Value (this report)	Feeds
k_{surf} (fitted, synthetic demo)	1.97×10^4 $\text{M}^{-1}\text{s}^{-1}$	LOD, ULOQ, classifier check
LOD / ULOQ	23.9 / 168.2 nM	Dynamic range, Sec. VII
Dynamic range	0.85 decades	Clinical validation
HNE bias (3% cross-react. placeholder)	$+3.0\%$ (chronic)	Sec. VII priority list
Matrix correction factor m	$1.19\times$	Clinical translation
MMP-9 decision boundary	13.4 nM	Sec. VII

IX. Limitations

1. k_{surf} **is not measured.** Every LOD, ULOQ, dynamic-range, and classifier-overlap number in this report is computed at a synthetic placeholder (2.0×10^4 $\text{M}^{-1}\text{s}^{-1}$) standing in for the as-yet-unmeasured fabricated-

electrode value. The confirmed solution-phase upper bound (67,500 $M^{-1}s^{-1}$, Kridel 2001) and Layer 9's 13,000–27,000 $M^{-1}s^{-1}$ target band bracket the plausible range, but only a real calibration run pins the value.

2. **A dimensional error in Layer 9's Eq. IV.5 was found and corrected here.** Section I documents the fix (dividing through by $K^{\Gamma}_{m,\mathrm{surf}}$ to recover a dimensionless equation in k_{surf} and ρ). This should be back-ported to Layer 9's own report and code so the two stay consistent.

3. **HNE cross-reactivity on SGKIPRTLTA is unconfirmed.** Section IV's ξ_{HNE} and bias numbers use a 3% placeholder with no wet-lab support (Layer 9, Table 3, flags this as "moderate risk"). The bias grows steeply with this fraction; measuring it is high priority.

4. ρ **(probe-saturation ratio) is inherited unconfirmed from Layer 9.** $\rho \approx 3.1$ depends on $K^{\Gamma}_{m,\mathrm{surf}}$ (via an unconfirmed δ_{eff}, Layer 8) and Γ_0 (literature midpoint, not the fabricated device's actual probe density).

5. **Wound-state uncertainties are approximate.** No paper propagates a formal uncertainty on $[\mathrm{MMP9}]_{\mathrm{free}}$; the symmetric $\sigma_{\log_{10}} = 0.15$ used in Section VI is a conservative stand-in, not a fitted value.

6. **The HNE-augmented classifier (Section VI) uses a single Gaussian for a bimodal quantity.** Chronic-wound HNE is elevated in only ∼50% of samples (Trengove 1999); a two-component mixture would be more faithful but was judged not worth the added complexity given HNE is not implemented on the real device.

7. **Matrix-effect and differential-fouling parameters are placeholders.** The 15% k_{surf} attenuation and 1.5% differential-fouling bias (Section V) have no wet-lab support; a real spike-and-recovery panel across the pH 6.5–8.0 range (`contents.md`, Sec. 15.1) is needed to replace them.

References

Papers confirmed in Layers 0, 9, and 13's own reference lists; drawn on directly here.

[1] Anne A, Demaille C (2012). Kinetics of enzyme action on surface-attached substrates: a practical guide to progress curve analysis in any kinetic situation. *Langmuir* 28:14665–14671. **anne_demaille2012.pdf** – exact surface-substrate ODE system and progress-curve solutions confirmed (Layer 9).

[2] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276(23):20572–20578. DOI: 10.1074/jbc.M100900200. **kridel2001.pdf** – $k_{\mathrm{cat}} = 86.4\ \mathrm{s}^{-1}$, $K_m = 1.28$ mM, $k_{\mathrm{cat}}/K_m = 67{,}500\ \mathrm{M}^{-1}\mathrm{s}^{-1}$ for SGKIPRTLTA (peptide A11) by MMP-9, Table II confirmed.

[3] Adjémian J, Anne A, Cauet G, Demaille C (2010). Cleavage-sensing peptide-enzyme multilayer gold surfaces for the parallel, selective, and sensitive assay of trypsin and plasmin activities by electrochemical methods. *Langmuir* 26:10347–10356. **adjemian2010.pdf** – experimental surface cleavage yields and Γ_0 context (Layer 9).

[4] Squires TM, Messinger RJ, Manalis SR (2008). Making it stick: convection, reaction and diffusion in surface-based biosensors. *Nat Biotechnol* 26:417–426. **squires2008.pdf** – Damköhler-number framework and surface MM ODE formulation (Layer 9).

[5] Pheeney CG, Barton JK (2012). DNA electrochemistry with tethered methylene blue. *Langmuir* 28:7063–7070. DOI: 10.1021/la300566x. **pheeney_barton2012.pdf** – MB surface coverage $\Gamma_0 = 1.3$–3.2 pmol cm^{-2} used for the ρ estimate (Layer 9, Table 1).

[6] White RJ, Plaxco KW (2010). Exploiting binding-induced changes in probe flexibility for the optimization of electrochemical biosensors. *Anal Chem* 82:73–76. DOI: 10.1021/ac9028595. **white2010.pdf** – ratiometric signal construction and electrode-area cancellation confirmed (Layer 13).

[7] Lubin AA, Plaxco KW (2010). Folding-based electrochemical biosensors: the case for responsive nucleic acid architectures. *Acc Chem Res* 43:496–505. DOI: 10.1021/ar900165x. **lubin_plaxco2010.pdf** – background-cleavage correction by control electrode and double-ratio fouling cancellation confirmed (Layer 13).

[8] Baker BR, Lai RY, Wood MS, Doctor EH, Heeger AJ, Plaxco KW (2006). An electronic, aptamer-based small-molecule sensor for the rapid, label-free detection of cocaine in adulterated samples and biological fluids. *J Am Chem Soc* 128:3138–3139. DOI: 10.1021/ja056957p. **baker2006.pdf** – on-chip reference stability and dynamic range in biological fluid confirmed (Layer 13).

[9] Lubin AA, Lai RY, Baker BR, Heeger AJ, Plaxco KW (2010). Electrochemical, structure-switching biosensors: quantitative analysis of the usable dynamic range. *Anal Chem* 78:5671–5677. DOI: 10.1021/ac0601417. **lubin2010.pdf** – per-electrode $R(0)$ baseline normalisation and dynamic range characteristics confirmed (Layer 13).

[10] Trengove NJ, Stacey MC, Macauley S, Bennett N, Gibson J, Burslem F, Murphy G, Schultz G (1999). Analysis of the acute and chronic wound environments: the role of proteases and their inhibitors. *Wound Repair Regen* 7(6):442–452. **trengove1999.pdf** – HNE elevated in only 6 of 13 chronic VLU samples; TIMP-1 values confirmed (Layer 0).

[11] Ladwig GP, Robson MC, Liu R, Kuhn MA, Muir DF, Schultz GS (2002). Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers. *Wound Repair Regen* 10(1):26–37. **ladwig2002.pdf** – chronic and healing active MMP-9 concentrations confirmed (Layer 0).

[12] International Conference on Harmonisation (2005). *ICH Harmonised Tripartite Guideline: Q2(R1), Validation of Analytical Procedures – Text and Methodology.* ICH, Geneva. **ich_q2r1_2005.pdf** – general LOD/LLOQ terminology referenced for comparison with Layer 13's own F_k-based detection thresholds (Section III); Layer 13's thresholds, not the generic $3\sigma/10\sigma$ blank-slope definition, are used for all numeric results in this report.

[13] Duda RO, Hart PE, Stork DG (2001). *Pattern Classification*, 2nd ed. Wiley, New York. **duda_hart_stork2001.pdf** – standard reference for multi-class Bayesian posterior classification (Section VI).